STRUCTURAL DYNAMICS

ENCE 365

Lecture	**: 3**	**Year : III**
Tutorial	**: 2**	**Part : II**
Practical	**: 1**	

Course Objectives:

The objective of this course is to equip students with an understanding of the basic concepts of structural dynamics and the dynamic behaviour of structures, along with the underlaying principles required to analyze and solve dynamic problems of structures.

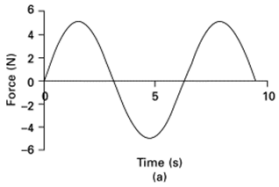
STRUCTURAL DYNAMICS

1.0 Introduction(6 hours)

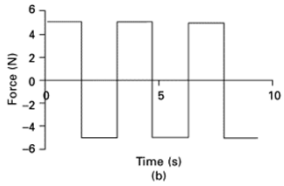
1.1 Structural dynamics (Dynamic loading, structural behavior and response)
1.2 Essential characteristics of a dynamic problem
1.3 Equation of motion (Newton's second law of motion and D'Alembert's principle) and dynamic degree of freedom
1.4 Types of vibrations (Free, forced, damped, undamped)
1.5 Types of dynamic loading (simple harmonic, periodic, transient, impulsive, random)
1.6 Damping: definition, characteristics and type (Viscous, Coulomb, structural, aerodynamic, magnetic, viscoelastic and active)
1.7 Dynamic response of structures
1.8 Time domain and frequency-domain analysis

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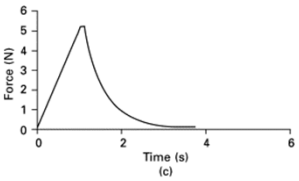
1.1 Structural Dynamics (Dynamic Loading)



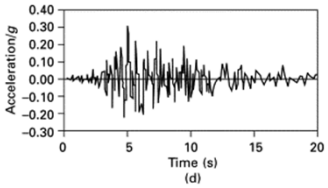
Periodic Harmonic



Periodic non-Harmonic



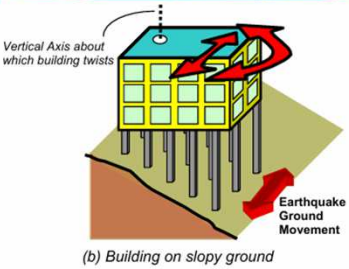
Ballast Load (Impulse)



Random Load (Earthquake)

1.1 Structural Dynamics (Structural Behavior)





(b) Building on slopy ground

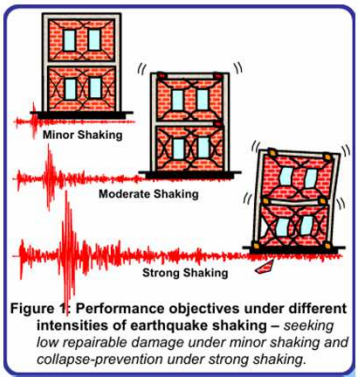
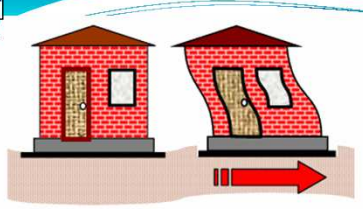
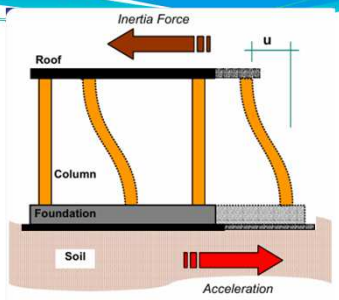
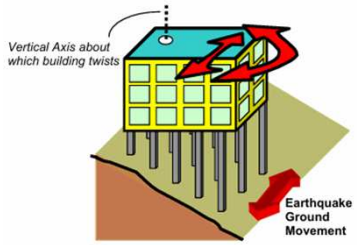


Figure 1: Performance objectives under different intensities of earthquake shaking – *seeking low repairable damage under minor shaking and collapse-prevention under strong shaking.*

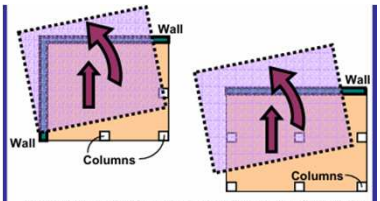
1.1 Structural Dynamics (Structural Response)







(b) Building on slopy ground



(c) Buildings with walls on two/one sides (in plan)

Figure 4: Buildings have unequal vertical members; they cause the building to twist about a vertical axis.

1.2 Essential Characteristics of dynamic problem

The **essential characteristics of a dynamic problem** :

Changes Over Time

The conditions, variables, or environment keep changing while the problem is being solved.

Interdependence of Variables

Different parts of the problem influence each other, so changing one factor may affect many others.

Feedback Effects

Actions taken to solve the problem can create new situations or consequences that influence future decisions.

Uncertainty and Risk

Future outcomes are not fully predictable because information may be incomplete or conditions may shift unexpectedly.

Need for Continuous Decision-Making

Solutions are not usually one-time fixes; they require monitoring, adaptation, and repeated decisions.

Multiple Possible Solutions

There may not be a single correct answer. Different strategies may work depending on changing circumstances.

Complexity

Dynamic problems usually involve many interacting elements, making analysis and prediction difficult.

Example

- Managing traffic in a city is a dynamic problem because:
- traffic volume changes constantly,
- accidents affect routes,
- weather changes conditions,
- and traffic-control decisions influence future congestion.

Newton's Second Law of Motion

It states that the acceleration of an object depends upon two variables – the net force acting on the object and the mass of the object. The acceleration of the body is directly proportional to the net force acting on the body and inversely proportional to the mass of the body. This means that as the force acting upon an object is increased, the acceleration of the object is increased. Likewise, as the mass of an object is increased, the acceleration of the object is decreased.



1.3 Equation of Motion (Newton's Second Law of Motion & D'Alembert's principle) and dynamic degree of freedom

Jean Le Rond d'Alembert, (born November 17, 1717, Paris, France—died October 29, 1783, Paris), French mathematician, philosopher, and writer, who achieved fame as a mathematician and scientist before acquiring a considerable reputation as a contributor to and editor of the famous *Encyclopédie*.



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D'Alembert's principle, alternative form of Newton's second law of motion:

In effect, the principle reduces a problem in dynamics to a problem in statics.

Second law: force *F* acting on a body is equal to the product of mass *m* & acceleration *a* of the body,

or $F = ma;$

In other words, the body is in equilibrium under the action of the real force *F* and the fictitious force *-ma*.

$F - ma = o.$

The fictitious force is also called an inertial force and a reversed effective force.

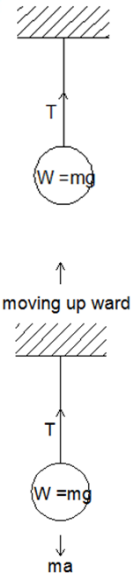
Assume weight 'w' suspended from a wire. The weight is subjected to gravitational force, w = mg. That is tension in wire, T = w = mg.

If mass accelerates upward with an acceleration 'a'. From Newton's second law, T-w = ma. Or T = w + ma.

The tension in the wire counteract an additional force ma = m ü(t), direction of force is opposite to acceleration.

In other words time derivative of momentum = force.

∴Dynamic loads are time varying loads. But time varying loads may not requires dynamic analysis.



1.3 Equation of Motion (Newton's Second Law of Motion & D'Alembert's principle) and dynamic degree of freedom

A static load is one which varies very slowly. A dynamic load is one which changes with time fairly quickly in comparison to the structure's natural frequency.

If a load is applied sufficiently slowly, the inertia forces (Newton's first law of motion) can be ignored and the analysis can be simplified as static analysis.

But if it varies quickly (relative to the structure's ability to respond), the response must be determined with a dynamic analysis.

Structural dynamics, therefore, is a type of structural analysis which covers the behavior of structures subjected to dynamic (actions having high acceleration) loading.

1.3 Equation of Motion (Newton's Second Law of Motion & D'Alembert's principle) and dynamic degree of freedom

Dynamic loads include people, wind, waves, traffic, earthquakes, and blasts. Any structure can be subjected to dynamic loading. Dynamic analysis can be used to find dynamic displacements, time history, and modal analysis.

A dynamic analysis is also related to the inertia forces developed by a structure when it is excited by means of dynamic loads applied suddenly (e.g., wind blasts, explosion, earthquake).

Dynamic analysis for simple structures can be carried out manually, but for complex structures finite element analysis can be used to calculate the mode shapes and frequencies.

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1.3 Equation of Motion (Newton's Second Law of Motion & D'Alembert's principle) and dynamic degree of freedom

Statics, dynamics and structural dynamics:

Statics deals with the effect of forces on bodies at rest.

Dynamics deals with the motion of nominally rigid bodies. The two aspects of dynamics are *kinematics* and *kinetics*.

Kinematics is concerned only with the motion of bodies with geometric constraints, irrespective of the forces acting on.

On the other hand, in kinetics, the path of a particle may vary as a result of the applied forces.

The term structural dynamics implies that, in addition to having motion, the bodies are non-rigid, i.e. 'elastic'.

1.3 Dynamic Degree of Freedom

The minimum number of independent coordinates required to determine completely the positions of parts of a system at any instant of time define the degree of freedom (DOF) of the system.

A large number of practical systems can be described using a finite number of DOFs.

Systems with a finite number of DOFs are called discrete or lumped parameter system.

Some systems, especially those involving continuous elastic members, have an infinite number of DOFs.

Those systems are called continuous or distributed systems.

1.3 Dynamic Degree of Freedom

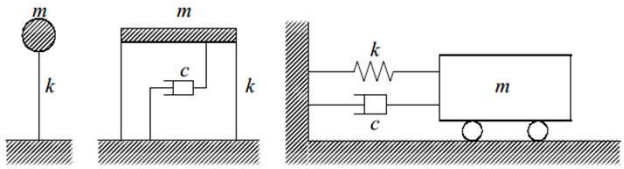
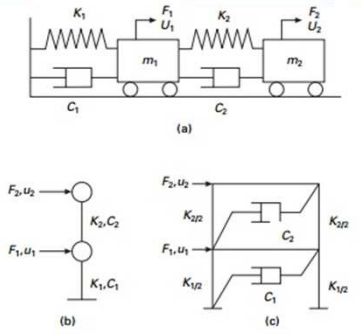
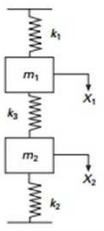


Fig.2.1 Analytical models of single degree of freedom (SDOF) systems

1.3 Dynamic Degree of Freedom

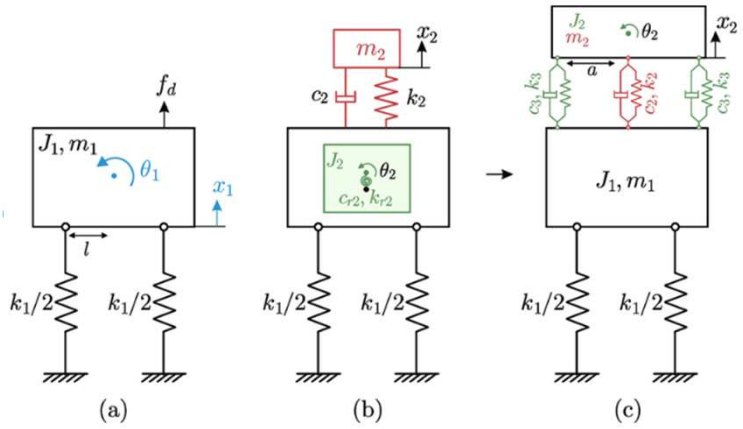


9.1 Representation of two degrees of freedom.

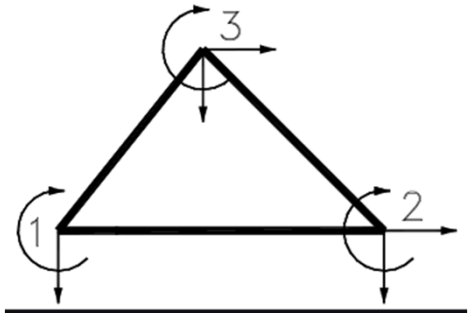


9.2 Two-degrees-of-freedom system.

1.3 Dynamic Degree of Freedom



1.3 Dynamic Degree of Freedom



1.4 Types of Vibrations (Free, Forced, Damped and undamped)

Any motion that repeats itself after an interval of time is called *vibration* or *oscillation*.

The general terminology of "Vibration" is used to describe oscillatory motion of mechanical and structural systems.

People became interested in vibration when the first musical instruments, probably whistles or drums, were discovered.

Most human activities involve vibration in one form or other. For example, we hear because our eardrums vibrate and see because light waves undergo vibration.

The Vibration of a system involves the transfer of its potential energy to kinetic energy and kinetic energy to potential energy, alternately.

1.4 Types of Vibrations (Free, Forced, Damped and undamped)

A vibratory system in general includes

- a means for storing potential energy (spring or elastic),
- a means for storing kinetic energy (mass or inertia) and
- a means by which energy is gradually lost (damper).

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1.4 Types of Vibrations (Free, Forced, Damped and undamped)

The terminology of "Free Vibration" is used for the study of natural vibration modes in the absence external loading.

The terminology of "Forced Vibration" is used for the study of motion as a result of loads that vary rapidly with time.

Loads that vary rapidly with time are called dynamic loads.

If no energy is lost or dissipated in friction or other resistance during oscillation, the vibration is known as "undamped vibration".

If any energy is lost in this way, however, is called "damped vibration".
If the system is damped, some energy is dissipated in each cycle of vibration and must be replaced by an external force if a state of steady vibration is to be maintained.

1.5 Types of Dynamic Loading (Simple Harmonic, Periodic, Transient, Impulsive and Random)

Dynamics loads can be grouped in one of the following categories (in order of increasing complexity):

- Simple Harmonic loads
- Periodic loads
- Transient loads
- Impulsive loads
- Random loads

1.5 Types of Dynamic Loading (Simple Harmonic)

Simple Harmonic loads:

Simple harmonic motion can be characterized by sinusoidal motion at constant frequency. Its most important features are: amplitude, frequency and phase. Simple harmonic motion can be described in different ways like:

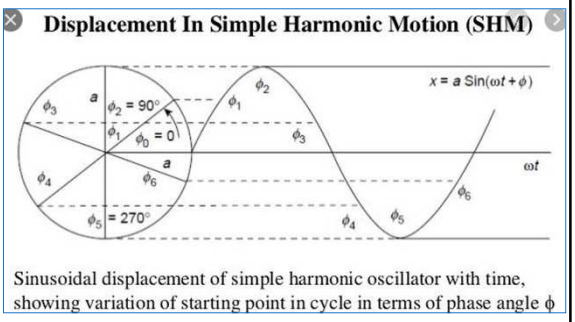
Trigonometric notation: $u(t) = A\,Sin(\omega t+\phi)$ ... 1

Where,

A = Amplitude

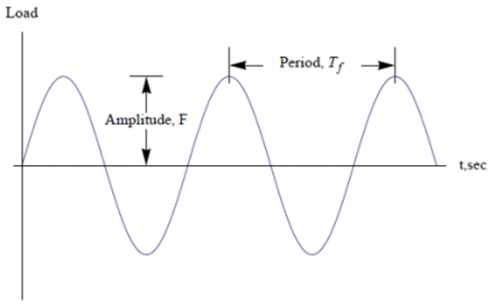
ω = Circular frequency

f = phase angle



Sinusoidal displacement of simple harmonic oscillator with time, showing variation of starting point in cycle in terms of phase angle ϕ

1.5 Types of Dynamic Loading (Simple Harmonic)



Complex Notation for simple harmonic load:

$u(t) = a(e^{i\omega t}+ e^{-i\omega t})/2-bi(e^{i\omega t}-e^{-i\omega t})/2$

Where,

a, b = Constants

ω = Circular frequency

1.5 Types of Dynamic Loading (Simple Harmonic)

Response to Simple Harmonic loads:

The response of a system to a harmonic excitation (loading) is called harmonic response.

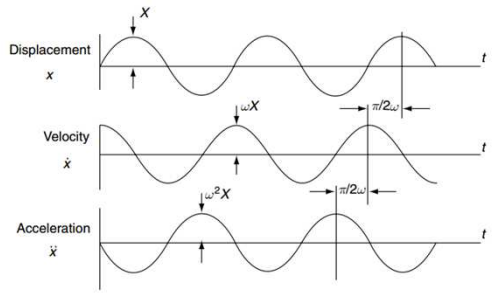


Fig. 1.2 Displacement, velocity and acceleration time histories for simple harmonic motion.

1.5 Types of Dynamic Loading (Simple Harmonic)

If a motion is repeated after equal intervals of time it is called periodic motion. The simplest type of periodic motion is harmonic motion.

It is described by sine or cosine functions.

$u(t) = A\ Sin(\omega t+\phi)$, if $\phi = o$, then $u(t) = A\ Sin(\omega t)$

And, $\acute{u}(t) = -\omega A\ Cos(\omega t)$

And, $\ddot{u}(t) = -\omega^2 A\ Sin(\omega t) = -\omega^2\ u(t)$

Where, ω = frequency (rad/sec)

A = Amplitude

1.5 Types of Dynamic Loading (Simple Harmonic)

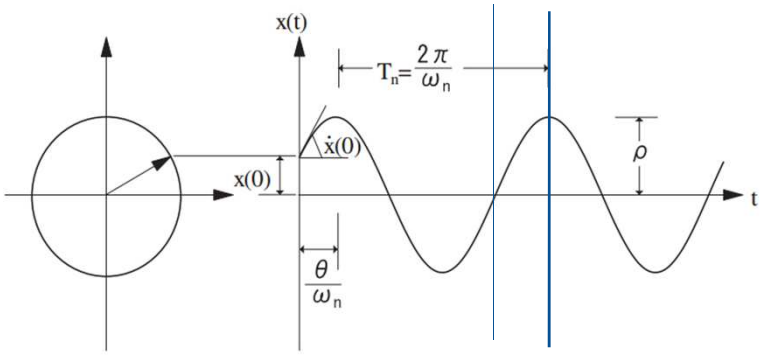
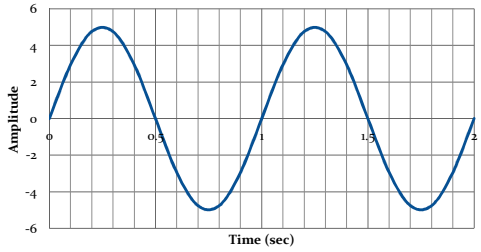


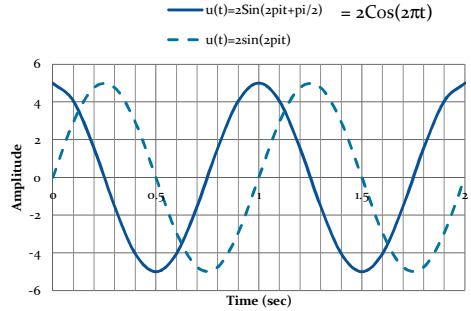
Fig.2.2.2 Undamped free-vibration response

Plot of $u(t) = \rho \, Sin(\omega t + \phi)$, where $\phi = \theta/\omega_n$

1.5 Types of Dynamic Loading (Simple Harmonic)

Plot of u(t)=2Sin(2πt)





1.5 Types of Dynamic Loading (Simple Harmonic)

Harmonic motion can be written in terms of exponential functions.

$$Sin(\omega t) = \frac{e^{i\omega t} - e^{-i\omega t}}{2i}; Cos(\omega t) = \frac{e^{i\omega t} + e^{-i\omega t}}{2i}$$

So that,

$$e^{i\omega t} = \text{Cos}\,(\omega t) + iSin(\omega t)$$

A harmonic motion could be written as,

$$u(t) = a.e^{i\omega t}$$

Generally a harmonic motion can be expressed as a combination of Sine & Cosine waves.

$$u(t) = ACos(\omega t) + BSin(\omega t) \quad \leftrightarrow \quad u(t) = YSin(\omega t + \theta)$$

Where,

$$Y = \sqrt{A^2 + B^2} \qquad \theta = tan^{-1}\left(A/B\right)$$

Or, $u(t) = ACos(\omega t) - BSin(\omega t) \quad \leftrightarrow \quad u(t) = -YSin(\omega t - \theta)$

$$Or,\ u(t) = YCos(\omega t - \theta)$$

1.5 Types of Dynamic Loading (Transient)

Transient loads:

A **transient load** is a **load** that comes and goes often quickly.

Traffic on a bridge, wind gusts, and water in a sewer bridge are all can be **transient loads**.

There are two types:

Impact : Equipment dropped on a structure (inelastic collision)

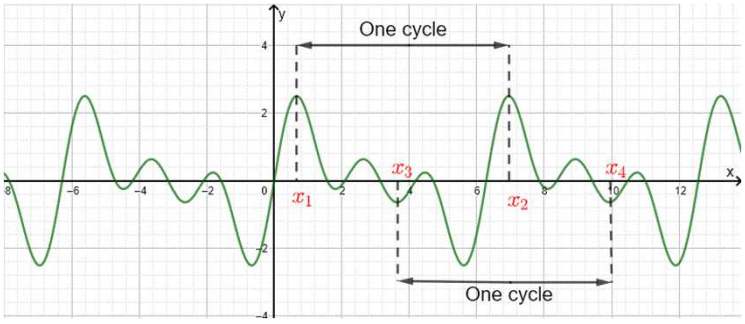
Blast load or impulse: Force impulse due to explosion.

1.5 Types of Dynamic Loading (Periodic)

Periodic loads:

A **periodic function** is a **function** that repeats its values at regular intervals.

Wave motion on off-shore structures.



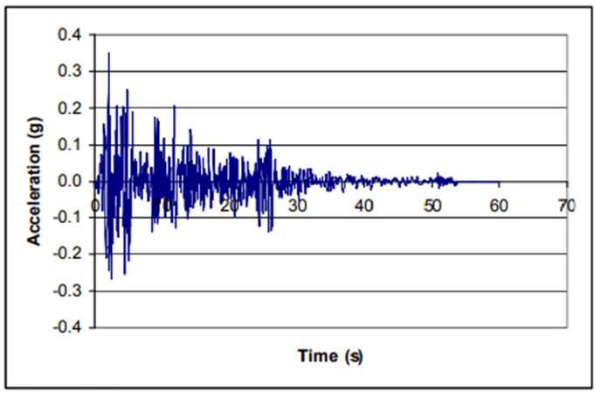
1.5 Types of Dynamic Loading (Random)

Random loads:

General arbitrary loads are complex and difficult to describe mathematically. As like:

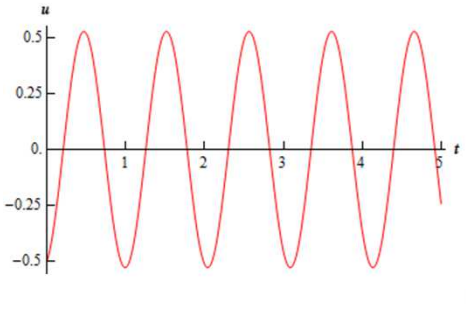
Earthquake load

Wind load

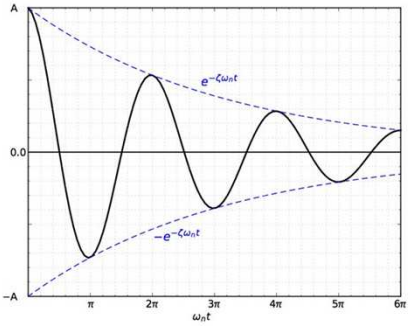


1.6 Damping: Definition, characteristics and type (Viscous, coulomb, structural, aerodynamic, magnetic, viscoelastic and active)

If no energy is lost or dissipated in friction or other resistance during oscillation, the vibration is known as “undamped vibration”. If any energy is lost in those way, is called “damped vibration”.



Undamped Vibration



Damped Vibration

1.6 Damping: Definition, characteristics and type (Viscous, coulomb, structural, aerodynamic, magnetic, viscoelastic and active)

All real structure will dissipate energy, mainly through friction.

Damping Vary from 2% to 10% generally.

Bolted steel structure ~ 6 %

RCC structure ~ 5 %

Welded Steel structure ~ 2 %

Brick Masonry ~ 10 %

A system may be so damped that it cannot vibrate.

Critical damping just prevents vibration or is just sufficient to allow the object to return to its rest position in the shortest period of time.

The automobile shock absorber is an example of a critically damped device.

1.6 Damping: Definition, characteristics and type (Viscous, coulomb, structural, aerodynamic, magnetic, viscoelastic and active)

Additional damping causes the system to be overdamped, which may be desirable, as in some door closers.

The vibrations of an under-damped system gradually taper off to zero.

There are many types of mechanical damping.

Friction, also called in this context dry, or Coulomb, damping, arises chiefly from the electrostatic forces of attraction between the sliding surfaces and converts mechanical energy of motion, or kinetic energy, into heat.

Viscous damping is caused by such energy losses as occur in liquid lubrication between moving parts or in a fluid forced through a small opening by a piston, as in automobile shock absorbers.

The viscous-damping force is directly proportional to the relative velocity between the two ends of the damping device.

1.6 Damping: Definition, characteristics and type (Viscous, coulomb, structural, aerodynamic, magnetic, viscoelastic and active)

Besides these external kinds of damping, there is energy loss within the moving structure itself that is called hysteresis damping or, sometimes, structural damping.

In hysteresis damping, some of the energy involved in the repetitive internal deformation and restoration to original shape.

Aerodynamic damping is the opposition to a moving or oscillating object caused by aerodynamic resistance (drag and lift forces). It extracts kinetic energy from vibrations, causing them to decay over time. This principle is critical for stabilizing everything from aircraft wings and wind turbines to tall buildings and rockets.

Positive Aerodynamic Damping: If the object's velocity or rotation causes a force that directly opposes its movement, the energy is dissipated, and the oscillation slows down.

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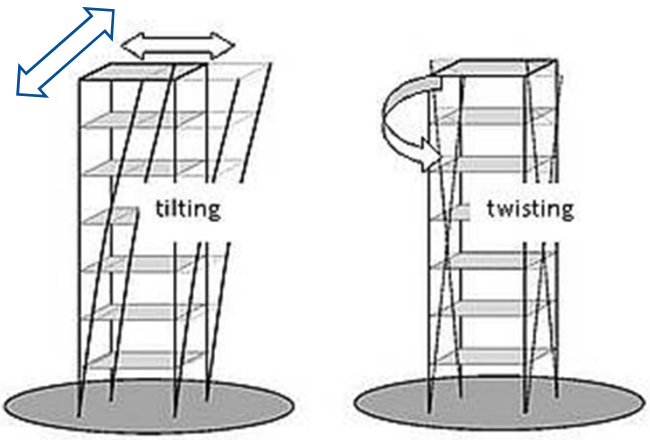
1.6 Damping: Definition, characteristics and type (Viscous, coulomb, structural, aerodynamic, magnetic, viscoelastic and active)

Negative Aerodynamic Damping: When an object's own motion generates a force that actually encourages or amplifies the movement, the system becomes unstable and can experience violent vibrations (e.g., flutter in wings).

Magnetic damping is a frictionless braking force that slows or stops the motion of a moving conductor using magnetic fields. As a conductor moves through a magnetic field, the changing magnetic flux generates circulating electric currents—known as eddy currents. These currents then create their own opposing magnetic field, which acts to resist the original motion.

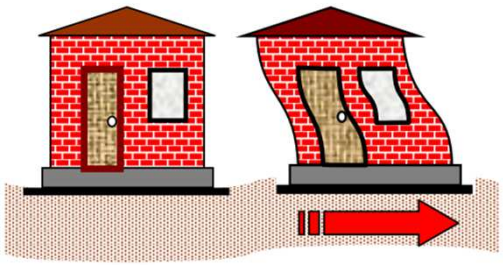
Active damping is a control system technique that uses sensors, algorithms, and actuators to automatically counteract, mitigate, or suppress unwanted vibrations and oscillations in real time

1.7 Dynamic Response of Structures

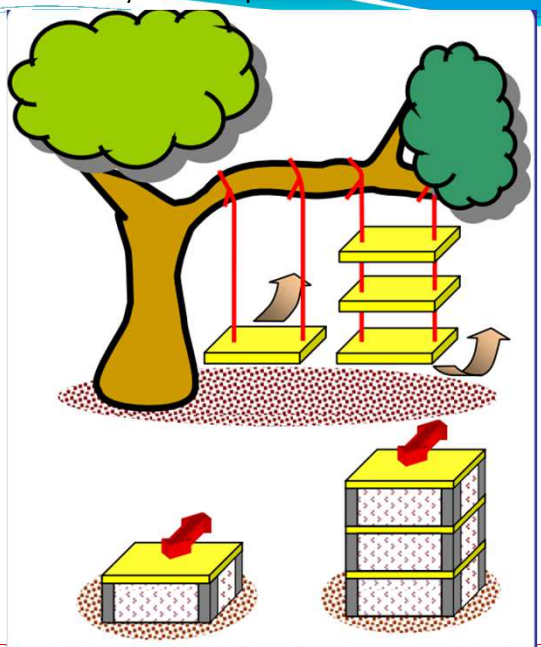


Longitudinal , Transverse and Torsional Vibration

1.7 Dynamic Response of Structures

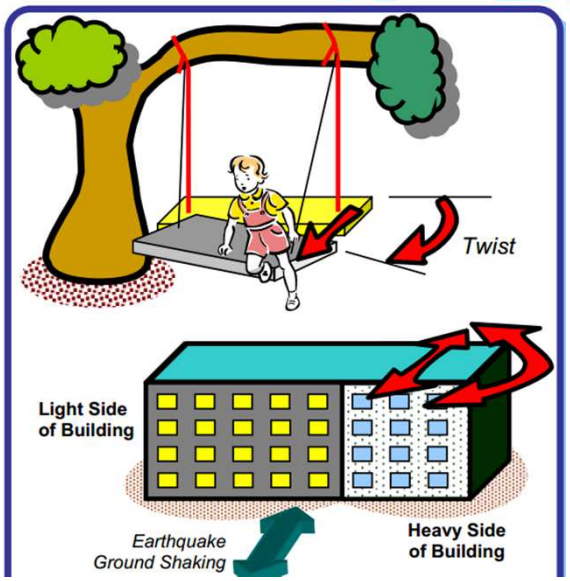


1.7 Dynamic Response of Structures



(a) Single-storey building (b) Three-storey building

1.7 Dynamic Response of Structures



1.7 Dynamic Response of Structures

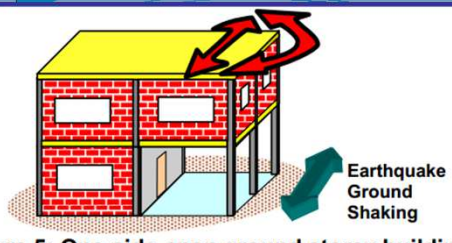
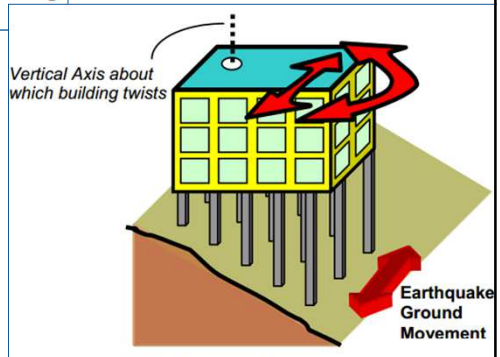


Figure 5: One-side open ground storey building twists during earthquake shaking.



(b) Building on slopy ground

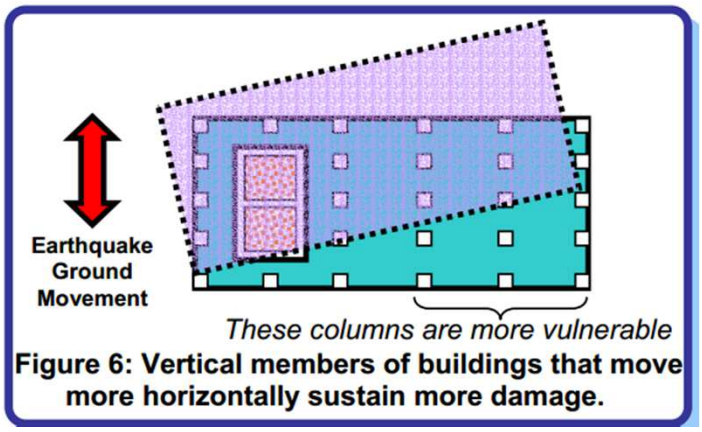
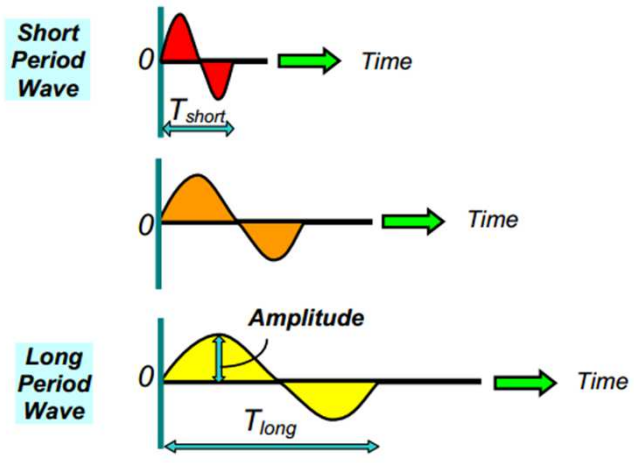
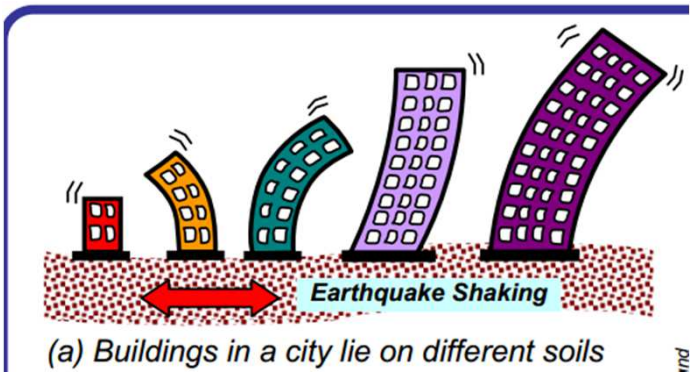


Figure 6: Vertical members of buildings that move more horizontally sustain more damage.



1.7 Dynamic Response of Structures



(a) Buildings in a city lie on different soils

1.8 Time-domain and frequency-domain analysis

The **time domain** and **frequency domain** are two different ways to analyze and represent the same signal or waveform.

The time domain shows how a signal's amplitude changes over time. It is the most intuitive perspective because it represents the "real world"—what happens sequentially, moment by moment.

$v = u + at = at + u$ ... 1 (v, t are variable, a & u constant)

$y = mx + c$ ... 2 (y, x are variable, m & c constant)

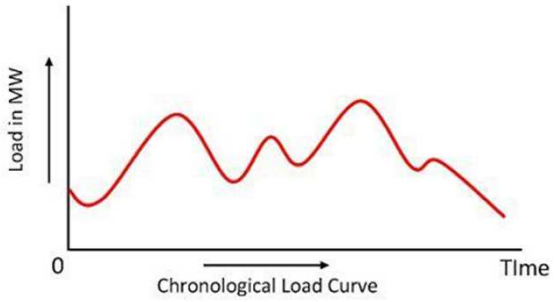
$v = at$ ... 3

$y = mx$ ... 4

Where, v = velocity, a = acceleration & t = time, similarly y = ordinate, m = slope and x = abscissa.

Visual Graph: Amplitude (e.g., voltage, sound pressure) on the vertical Y-axis, and Time (e.g., seconds) on the horizontal X-axis.

Best Used For: Tracking signal changes over time, identifying sudden spikes (transients), and evaluating overall signal integrity (checking for distortion, noise, or timing errors).



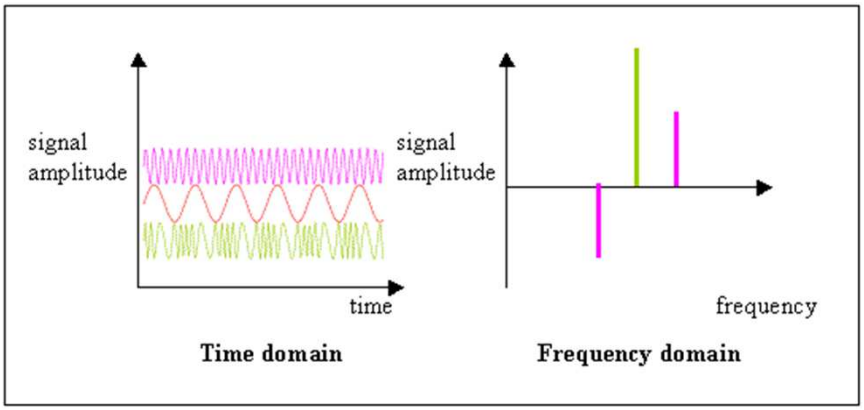
Frequency domain is an analysis of signals or mathematical functions, in reference to frequency, instead of time.

Frequency domain displays how much of the signal exists within a given frequency band concerning a range of frequencies. Also, a frequency-domain representation can include information on the phase shift that must be applied to each sinusoid to be able to recombine the frequency components to recover the original time signal.

Visual Graph: Amplitude on the vertical Y-axis, and Frequency (Hertz) on the horizontal X-axis.

Best Used For: Analyzing the specific harmonic content of a signal, filtering out unwanted noise, and evaluating the frequency response of circuits or audio equipment.

1.8 Time-domain and frequency-domain analysis



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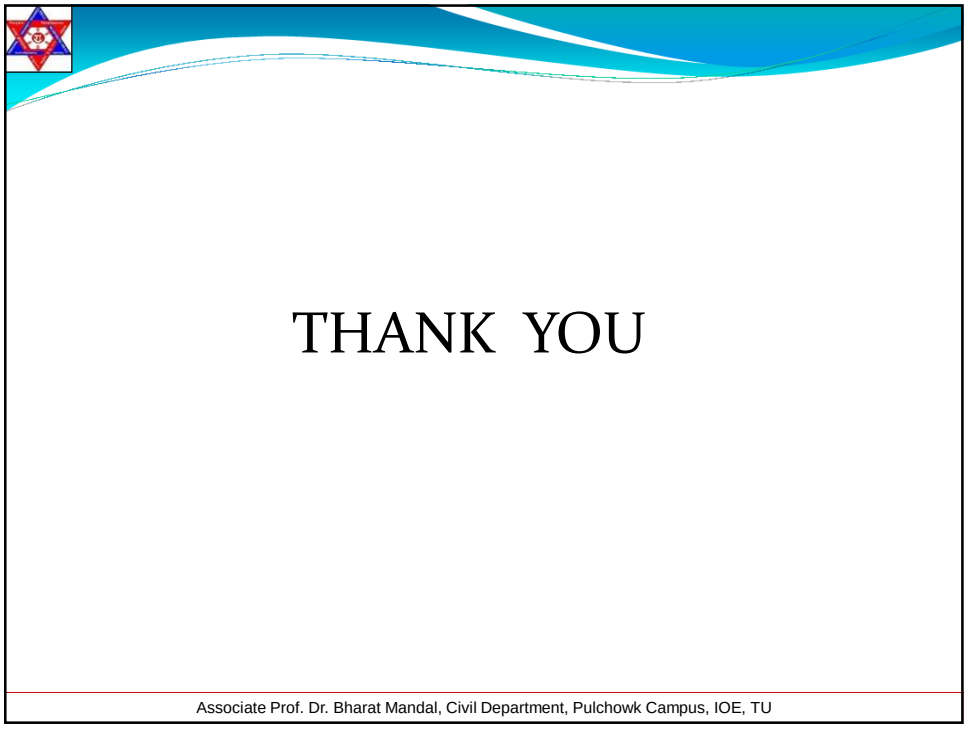
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THANK YOU
Associate Prof. Dr. Bharat Mandal, Civil Department, Pulchowk Campus, IOE, TU